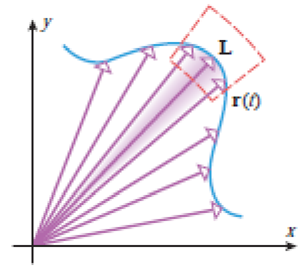
MTH 202 Chapter 2: Vector-Valued Functions

2.2 Calculus of Vector-Valued Functions

<u>Recall:</u> A vector-valued function $\mathbf{r}(t)$ maps real numbers to n-dimensional vectors $\mathbf{r}:\mathbb{R}\to\mathbb{R}^n$.

It is important to visualize how $\mathbf{r}(t)$ plots points via vectors.
We begin this section by looking at the limit of a vector-valued function.



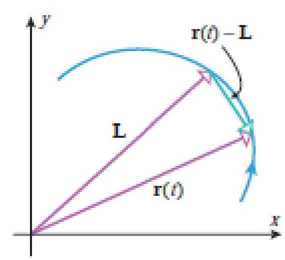
Definition: Let $\mathbf{r}(t)$ be a vector-valued function that is defined for all t in some open interval containing the number a (it need not be defined at a). We write that

$$\lim_{t\to a}\mathbf{r}(t)=\mathbf{L}$$

if and only if

$$\lim_{t\to a}\left\|\mathbf{r}(t)-\mathbf{L}\right\|=\mathbf{0}$$

Note: It is important to see that the limit of $\mathbf{r}(t)$ *is a vector* $\mathbf{L}$. $\mathbf{L}$ *is often referred to as a <u>limiting vector</u>. Therefore, we don't imagine we following points on the curve when we evaluate the limit of* $\mathbf{r}(t)$, *but rather we are following the sequence of vectors since* $\mathbf{r}(t)$ *is a function that produces vectors.*



The above definition suggests the following theorem.

Theorem: If $\mathbf{r}(t)=\langle f(t),g(t),h(t)\rangle$, then we define $\lim_{t\to a}\mathbf{r}(t)$ as follows:

$$\lim_{t\to a}\mathbf{r}(t)=\left\langle \lim_{t\to a}f(t),\lim_{t\to a}g(t),\lim_{t\to a}h(t)\right\rangle.$$

Proof: Omitted, but it follows from the above definition.

Definition: A vector function $\mathbf{r}(t)$ is <u>continuous at a</u> if $\lim_{t\to a}\mathbf{r}(t)=\mathbf{r}(a)$.

Theorem: A vector function **r** is continuous at a if and only if its component functions are continuous at a.

Proof: This theorem follows immediately from how continuity is defined for vector functions and how continuity is defined in calculus one for functions of the form $f:\mathbb{R}\to\mathbb{R}$.

Example 1: Consider the vector function $\mathbf{r}(t)=\left(\frac{t^2-t}{t-1}\right)\mathbf{i}+\left(\sqrt{t+8}\right)\mathbf{j}+\left(\frac{\sin(\pi t)}{\ln(t)}\right)\mathbf{k}$.

a) Find $\lim\limits_{t\to 1}\mathbf{r}(t)$

We use definition above the properties of limits from calculus one and l'Hopital's rule to discover:

$$\lim_{t\to 1}\mathbf{r}(t)=\left(\lim_{t\to 1}\frac{t^2-t}{t-1}\right)\mathbf{i}+\left(\lim_{t\to 1}\sqrt{t+8}\right)\mathbf{j}+\left(\lim_{t\to 1}\frac{\sin(\pi t)}{\ln(t)}\right)\mathbf{k}\overset{H}{=}\left(\lim_{t\to 1}t\right)\mathbf{i}+\left(\sqrt{9}\right)\mathbf{j}+\left(\lim_{t\to 1}\frac{\pi\cos(\pi t)}{(1/t)}\right)\mathbf{k}$$

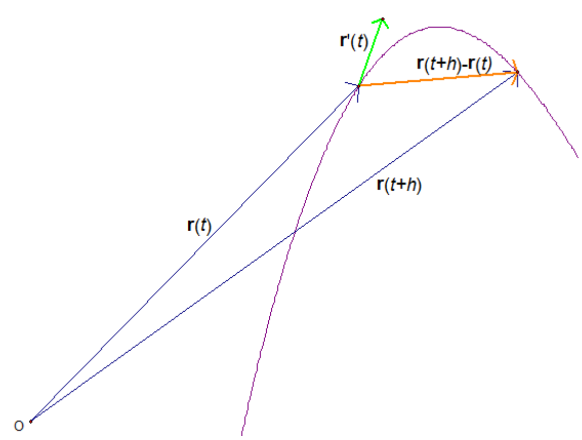
$$=(1)\mathbf{i}+(3)\mathbf{j}+(-\pi)\mathbf{k}=\mathbf{i}+3\mathbf{j}-\pi\mathbf{k}.$$

b) Is the function **r** continuous at 1? Give a brief explanation of your answer.

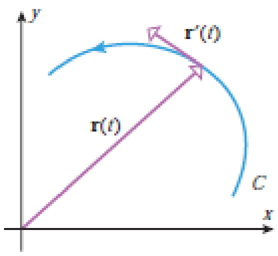
We note that $\mathbf{r}(1)$ is not defined. Thus, $\lim\limits_{t\to 1}\mathbf{r}(t)\neq\mathbf{r}(1)$, and **r** is not continuous at 1 by definition.

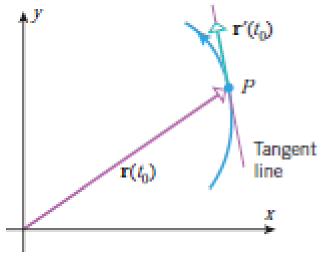
Definition: The <u>derivative</u> of a vector function **r** is defined by: $\frac{d\mathbf{r}}{dt}=\mathbf{r}'(t)=\lim\limits_{h\to o}\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$.

Intuitively, one can imagine that as h approaches 0 the vector in orange will approach the vector in green (which is tangent to the curve at the terminal point of $\mathbf{r}(t)$). This leads naturally to the following definitions.



Definition: Suppose that $\mathbf{r}'(t) \neq \mathbf{0}$. The vector $\mathbf{r}'(t)$ is called the <u>tangent vector</u> to the curve, C, defined by $\mathbf{r}$ at the terminal point, P, of the position vector $\mathbf{r}(t)$.
The <u>tangent line</u> to C at P is defined to be the line through P parallel to $\mathbf{r}'(t)$.
Also, $\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ is called the <u>unit tangent vector</u>.





The next theorem gives us a familiar way to calculate $\mathbf{r}'(t)$.

Theorem: If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ and the component functions of $\mathbf{r}$ are differentiable then, $\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$.

Proof:

Using the definition of $\mathbf{r}'(t)$ we calculate:

$$\begin{aligned}
\mathbf{r}'(t) &= \lim_{k\to 0} \frac{\mathbf{r}(t+k) - \mathbf{r}(t)}{k} = \lim_{k\to 0} \frac{1}{k}\Big(\langle f(t+k), g(t+k), h(t+k)\rangle - \langle f(t), g(t), h(t)\rangle\Big) \\
&= \lim_{k\to 0} \frac{1}{k}\Big(\langle f(t+k) - f(t), g(t+k) - g(t), h(t+k) - h(t)\rangle\Big) \\
&= \lim_{k\to 0}\left(\left\langle \frac{f(t+k)-f(t)}{k}, \frac{g(t+k)-g(t)}{k}, \frac{h(t+k)-h(t)}{k} \right\rangle\right) \\
&= \left\langle \lim_{k\to 0}\frac{f(t+k)-f(t)}{k}, \lim_{k\to 0}\frac{g(t+k)-g(t)}{k}, \lim_{k\to 0}\frac{h(t+k)-h(t)}{k} \right\rangle.
\end{aligned}$$

Since the component functions of $\mathbf{r}$ are differentiable we know

$$\left\langle \lim_{k\to 0}\frac{f(t+k)-f(t)}{k}, \lim_{k\to 0}\frac{g(t+k)-g(t)}{k}, \lim_{k\to 0}\frac{h(t+k)-h(t)}{k} \right\rangle = \langle f'(t), g'(t), h'(t) \rangle.$$

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Note: There is nothing special in the above proof that requires the vector function to have 3 components. The same proof works for a vector function of the form $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$.

Example 2: Consider $\mathbf{r}(t)=\langle 2\cos(t),\sin(t),t\rangle$

a) Find $\mathbf{r}'(t)$

We calculate: $\mathbf{r}'(t)=\langle -2\sin(t),\cos(t),1\rangle$.

b) Find $\mathbf{T}(t)$

We calculate

$\mathbf{T}(t)=\dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}=\dfrac{1}{\sqrt{4\sin^2(t)+\cos^2(t)+1}}\langle -2\sin(t),\cos(t),1\rangle=\dfrac{1}{\sqrt{3\sin^2(t)+2}}\langle -2\sin(t),\cos(t),1\rangle$ provided $\mathbf{r}'(t)\neq\mathbf{0}$.

c) Find parametric equations for the tangent line to the curve determined by $\mathbf{r}$ at the point $\left(0,1,\frac{\pi}{2}\right)$.

A direction vector parallel to the desired line is $\mathbf{r}'\left(\frac{\pi}{2}\right)=\langle -2,0,1\rangle$. So, a vector equation for the line is $\langle x,y,z\rangle=\left\langle 0,1,\frac{\pi}{2}\right\rangle+t\langle -2,0,1\rangle$, and parametric equations for the line are $x=-2t$ $y=1$ $z=\frac{\pi}{2}+t$.

Note: You can use the graphing software on Blackboard to help visualize what we have found in this example.

Theorem: Suppose $\mathbf{u}:\mathbb{R}\to\mathbb{R}^3$ and $\mathbf{v}:\mathbb{R}\to\mathbb{R}^3$ are differentiable vector functions, c is a scalar, and $f:\mathbb{R}\to\mathbb{R}$ is a differentiable function. Then,

1. $\dfrac{d}{dt}\left[\mathbf{u}(t)+\mathbf{v}(t)\right]=\mathbf{u}'(t)+\mathbf{v}'(t)$
2. $\dfrac{d}{dt}\left[c\mathbf{u}(t)\right]=c\mathbf{u}'(t)$
3. $\dfrac{d}{dt}\left[f(t)\mathbf{u}(t)\right]=f'(t)\mathbf{u}(t)+f(t)\mathbf{u}'(t)$
4. $\dfrac{d}{dt}\left[\mathbf{u}(t)\cdot\mathbf{v}(t)\right]=\mathbf{u}'(t)\cdot\mathbf{v}(t)+\mathbf{u}(t)\cdot\mathbf{v}'(t)$
5. $\dfrac{d}{dt}\left[\mathbf{u}(t)\times\mathbf{v}(t)\right]=\mathbf{u}'(t)\times\mathbf{v}(t)+\mathbf{u}(t)\times\mathbf{v}'(t)$
6. $\dfrac{d}{dt}\left[\mathbf{u}(f(t))\right]=f'(t)\mathbf{u}'(f(t))$

Proof: Many of these are left as practice in the exercises for this section.

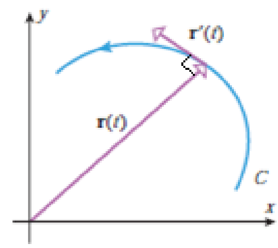
*Note: With the exception of property 5 all of the above properties hold when **u** and **v** have two or more component functions.*

Theorem: If $\mathbf{r}(t)$ is a differentiable vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t, then

$$\mathbf{r}(t)\cdot\mathbf{r}'(t)=0$$

that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t.

Proof:



By property 4 in the theorem above we know that:

$$\frac{d}{dt}\left[\mathbf{r}(t)\cdot\mathbf{r}(t)\right]=\mathbf{r}'(t)\cdot\mathbf{r}(t)+\mathbf{r}(t)\cdot\mathbf{r}'(t)$$

$$\Rightarrow\quad \frac{d}{dt}\left[\|\mathbf{r}(t)\|^2\right]=2\left[\mathbf{r}'(t)\cdot\mathbf{r}(t)\right]$$

However, since $\|\mathbf{r}(t)\|^2$ is a constant its derivative is 0 and therefore,

$$0=2\left[\mathbf{r}'(t)\cdot\mathbf{r}(t)\right]$$

$$\Rightarrow\quad 0=\mathbf{r}'(t)\cdot\mathbf{r}(t)$$

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Definition: The <u>definite integral</u> of a continuous vector function $\mathbf{r}(t)=\langle f(t),g(t),h(t)\rangle$ is $\int_a^b\mathbf{r}(t)dt=\left\langle\int_a^b f(t)dt,\int_a^b g(t)dt,\int_a^b h(t)dt\right\rangle$.

Note: For now we will hold off on a physical interpretation of the definite integral (we will see a nice application for it in 2.6). The indefinite integral is defined similarly.

<u>Example 3:</u> Suppose $\mathbf{r}(t)=(\cos(t))\mathbf{i}+\left(\frac{1}{1+t^2}\right)\mathbf{j}+\left(\sin^2(t)\right)\mathbf{k}$. Find $\int_0^{\frac{\pi}{4}}\mathbf{r}(t)\,dt$ (exactly).

We calculate

$$\int_0^{\frac{\pi}{4}}\mathbf{r}(t)\,dt=\left(\int_0^{\pi/4}\cos(t)\,dt\right)\mathbf{i}+\left(\int_0^{\pi/4}\frac{1}{1+t^2}\,dt\right)\mathbf{j}+\left(\int_0^{\pi/4}\sin^2(t)\,dt\right)\mathbf{k}$$

$$=\left(\sin(t)\Big|_0^{\pi/4}\right)\mathbf{i}+\left(\tan^{-1}(t)\Big|_0^{\pi/4}\right)\mathbf{j}+\left(\frac{1}{2}\int_0^{\pi/4}\left(1-\cos(2t)\right)dt\right)\mathbf{k}=\frac{\sqrt{2}}{2}\mathbf{i}+\tan^{-1}\left(\frac{\pi}{4}\right)\mathbf{j}+\left(\frac{1}{2}\left(t-\frac{1}{2}\sin(2t)\right)\Big|_0^{\pi/4}\right)\mathbf{k}$$

$$=\frac{\sqrt{2}}{2}\mathbf{i}+\tan^{-1}\left(\frac{\pi}{4}\right)\mathbf{j}+\left(\frac{\pi}{8}-\frac{1}{4}\right)\mathbf{k}$$

$$=\frac{\sqrt{2}}{2}\mathbf{i}+\mathbf{j}+\left(\frac{\pi}{8}-\frac{1}{4}\right)\mathbf{k}$$

Note: Again, it is important to see that our result is a <u>vector</u>.